Market Regime Detection in Indian Equities Using Hidden Markov Models

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March 20, 2026

1 Abstract

This project investigates if the Hidden Markov Model (HMM) can identify market regimes in NSE-listed equities. Using daily price data for `RELIANCE.NS` from Yahoo Finance, the study constructs a feature set based on log returns, rolling volatility, and Relative Strength Index (RSI). A Gaussian HMM is fit to the feature space to infer hidden states that are interpreted as market regimes. The analysis includes both static in-sample exploration and a walk-forward framework to evaluate regime detection accurately. Two trading overlays are tested: a probability-weighted strategy and a threshold-based allocation rule. Results show that the HMM is able to produce interpretable regimes, but trading performance is mixed. While some configurations improve gross Sharpe ratios, the edge deteriorates in others. The findings suggest that regime detection is useful as a risk-filtering tool, though it cannot be used as a standalone trading strategy.

2 Introduction

2.1 Motivation

A market regime identifying framework attempts to classify market behavior into a small number of recurring states. These states are not directly observed; instead, they are inferred from observable market variables. HMMs are well suited to this task because they explicitly model a state process and allow the observed feature distribution to vary across states.

2.2 Problem Statement

The core questions addressed in this project are:

1. Can a Gaussian Hidden Markov Model identify economically meaningful market regimes in an Indian stock?
2. Are the inferred regimes stable enough to be used in a walk-forward setting?
3. Can regime probabilities support a simple allocation rule that improves risk-adjusted performance?
4. How sensitive are results to training window choice and transaction costs?
5. Can this work as a standalone strategy for swing-trades?

2.3 Prior Work

Hamilton (1989) formalized Markov-switching models for macroeconomic time series, showing how latent regimes can explain structural changes in observed data. In financial applications, HMMs and related state-space models have been used to classify bull/bear markets, volatility states, and credit cycles.

At the same time, financial regime detection faces persistent challenges. Regimes inferred in-sample can appear cleaner than they are out-of-sample. State probabilities can be statistically sharp without necessarily translating into profitable trading signals. Frequent state switching can lead to high turnover and taxes. This project is therefore framed not only as a predictive exercise, but also as an empirical test of whether regimes by themselves are actionable through a strategy or not.

3 Data and Features

3.1 Data Source and Sample Period

- **Asset:** `RELIANCE.NS`
- **Market:** NSEI
- **Source:** Yahoo Finance via `yfinance`
- **Sample Period:** January 1, 2015 to March 19, 2026
- **Frequency:** Daily adjusted closing prices

The project begins with a single-stock analysis to maintain interpretability and to test the workflow before generalizing to a larger cross-section.

3.2 Feature Set

The feature matrix is designed to capture three distinct aspects of market behavior:

1. **Log Returns**
$$r_t = 100 \cdot \log\left(\frac{P_t}{P_{t-1}}\right)$$
Returns capture short-horizon directional movement.

2. **Rolling Volatility**
$$\sigma_t = \mathrm{std}(r_{t-w+1}, \ldots, r_t), \quad w = 20$$
Volatility captures recent instability in returns.

3. **Relative Strength Index (RSI)** A 14-day RSI is used to measure short-term momentum and overbought/oversold behavior.

3.3 Feature Configuration

Feature	Window	Purpose
Log return	1 day	Directional price movement
Rolling volatility	20 days	Short-term risk / instability
RSI	14 days	Momentum / reversal signal

Table 1: Feature construction used in the project.

3.4 Data Handling

The feature pipeline applies the following preprocessing:

- Sort index chronologically
- Forward-fill missing prices when necessary
- Drop remaining missing values
- Align RSI to the return index
- Remove rows with insufficient rolling history

This ensures that the HMM is fit on a clean, synchronized feature matrix.

3.5 Descriptive Statistics

Descriptive statistics for the feature matrix generated from `RELIANCE.NS` over the sample period are shown below.

Variable	Mean	Std Dev	Min	Max
Return	0.0712	1.6990	-14.1032	13.7307
Volatility	1.5529	0.6933	0.5570	7.7406
RSI	52.9987	16.7960	7.7226	97.5634

Table 2: Descriptive statistics of the feature matrix.

3.6 Figures

The following figures are generated by the walk-forward script and are embedded directly in the report.

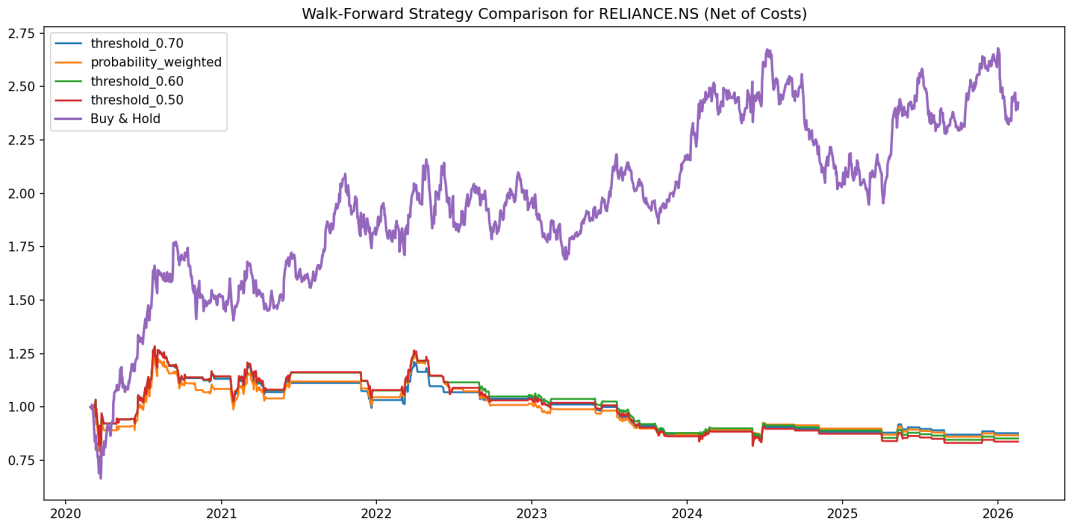

Figure 1: Equity curve comparison: strategy vs. buy-and-hold.

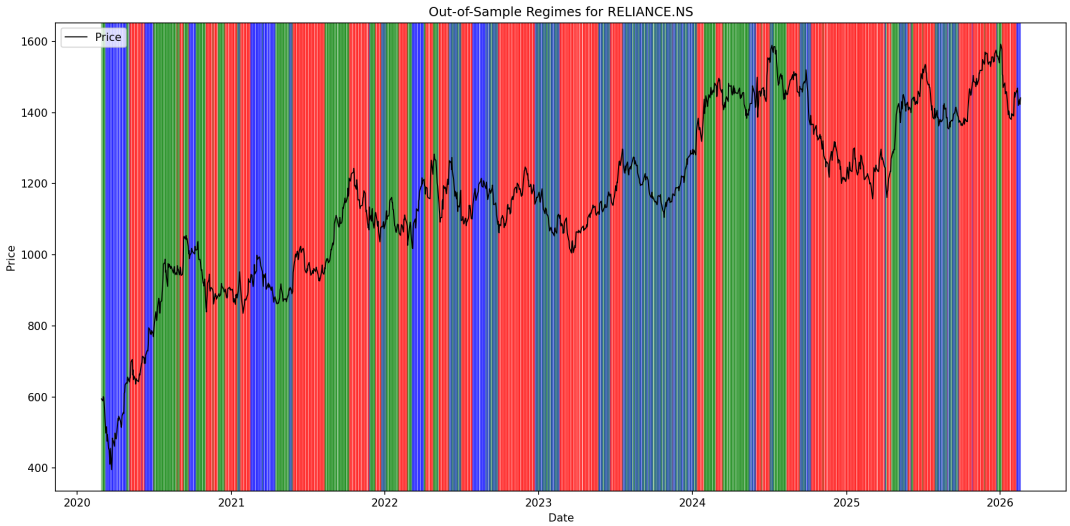

Figure 2: Historical regime classification (out-of-sample).

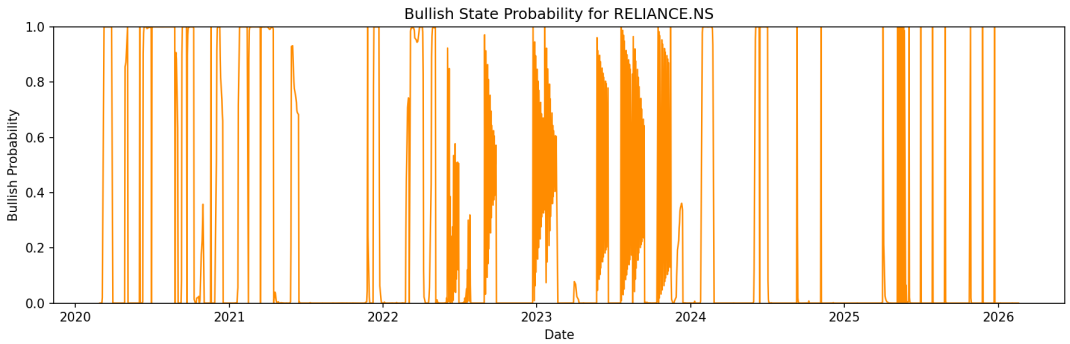

Figure 3: Posterior probability of the bullish state over time.

4 Methodology

4.1 Hidden Markov Model Specification

The project uses a Gaussian Hidden Markov Model with:

- Number of states: 3
- Covariance type: full
- Maximum iterations: 1000
- Random seed: 42

The model assumes that the observed feature vector X_t is generated conditionally on an unobserved state S_t, where the state sequence follows a first-order Markov chain.

4.2 Regime Interpretation

States are interpreted using state-conditional feature means. In particular, the state with the highest mean return is treated as the most bullish. States are then compared using volatility and RSI profiles to support economic interpretation.

4.3 Walk-Forward Design

The walk-forward framework uses rolling train/test windows.

- Default training window: 1250 trading days
- Test window: 20 trading days
- Sweep windows: 750, 1000, 1250, 1500

For each step:

1. Fit the HMM on the training slice
2. Identify the bullish state using mean returns
3. Compute regime probabilities on the next test slice
4. Store out-of-sample state assignments and probabilities

4.4 Trading Strategies

4.4.1 Probability-Weighted Strategy

The allocation weight equals the previous day's bullish-state posterior probability:

$$w_t = P(S_{t-1} = s_{\text{bull}} \mid X_{1:t-1})$$

4.4.2 Threshold Strategy

The allocation is either 0 or 1:

$$w_t = \begin{cases} 1 & \text{if } P(S_{t-1} = s_{\text{bull}}) \geq \theta \\ 0 & \text{otherwise} \end{cases}$$

Thresholds tested:

$$\theta \in \{0.50, 0.60, 0.70\}$$

4.5 Transaction Costs

A transaction cost of 10 basis points per unit of turnover is applied:

$$\text{turnover}_t = |w_t - w_{t-1}|$$

$$\text{cost}_t = \text{turnover}_t \times \frac{10}{10000}$$

5 Results

5.1 Baseline Walk-Forward Result

Strategy	Cum. Return	Sharpe	Buy & Hold Return	Buy & Hold Sharpe
Probability-weighted	4.59%	0.129	142.49%	0.676
Threshold 0.50	5.95%	0.143	142.49%	0.676
Threshold 0.60	6.53%	0.148	142.49%	0.676
Threshold 0.70	6.55%	0.148	142.49%	0.676

Table 3: Baseline walk-forward results for train window 1250 and test window 20.

5.2 Training Window Sweep

Strategy	Train Window	Threshold	Gross Return	Gross Sharpe	Buy & Hold Return	B
Threshold	1000	0.60	67.65%	0.639	156.93%	
Threshold	1000	0.50	61.87%	0.597	156.93%	
Threshold	1000	0.70	58.30%	0.584	156.93%	
Probability-weighted	1000	–	53.36%	0.549	156.93%	

Table 4: Best configurations from the train-window sweep.

5.3 Net-of-Cost Results

Strategy	Gross Return	Net Return	Gross Sharpe	Net Sharpe	Turnover	Total Cost
Threshold 0.60	67.65%	36.22%	0.639	0.409	208.00	20.80%
Probability-weighted	53.37%	27.46%	0.549	0.339	185.19	18.52%
Buy & Hold	156.93%	156.93%	0.628	0.628	n/a	n/a

Table 5: Net-of-cost performance for the best gross setup after applying 10 bps transaction costs.

6 Discussion

The results suggest that the HMM produces meaningful statistical regimes, but these regimes do not automatically generate a strong investment strategy.

The main issues are:

1. **High turnover**: daily updates to exposure generate large cumulative turnover.
2. **Signal fragility**: results are sensitive to the training window.
3. **Sharp posteriors do not imply tradable edge**: a model can be highly confident internally without adding economic value.
4. **Single-stock limitation**: results may reflect stock-specific dynamics rather than market-wide regimes.
5. **Limited feature set**: returns, volatility, and RSI are informative but not exhaustive.

The project therefore supports the use of HMMs as a descriptive and risk-monitoring tool more than as a standalone trading engine.

7 Conclusion and Future Work

This project developed a complete workflow for market regime detection in an Indian stock using a Gaussian Hidden Markov Model trained on returns, volatility, and RSI. The model produces interpretable latent states and the walk-forward framework reveals that performance is highly dependent on training-window design. While some gross Sharpe improvements were observed, much of the edge disappeared after transaction costs were introduced.

Future work should focus on:

- richer momentum and trend features
- turnover reduction via smoothing or holding constraints
- multi-stock and index-level analysis
- model selection criteria for state count
- broader dashboard functionality for regime monitoring

8 References

[1] Hamilton, J. D. (1989). *A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle.* Econometrica, 57(2), 357–384.

[2] Rabiner, L. R. (1989). *A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition.* Proceedings of the IEEE, 77(2), 257–286.

[3] Elliott, R., Aggoun, L., & Moore, J. (1995). *Hidden Markov Models: Estimation and Control.* Springer.

[4] Ang, A., & Bekaert, G. (2002). *International Asset Allocation with Regime Shifts.* Review of Financial Studies, 15(4), 1137–1187.

[5] Guidolin, M. (2011). *Markov Switching Models in Empirical Finance.* In Missing Data Methods: Time-Series Methods and Applications.

[6] Nystrup, P., Hansen, B. W., Larsen, H. O., & Lindström, E. (2016). *Detecting Change Points in VIX and S&P 500: A Hidden Markov Model Approach.* Journal of Asset Management, 17, 361–378.